

## Functions - Solutions

**Problem 1.** Find all functions  $f : \mathbb{R} \rightarrow \mathbb{R}$  such that

$$f(x)f(y) = f(x+y) + xy + x + y \quad (1)$$

for any  $x, y \in \mathbb{R}$ .

**Solution.** Putting  $y = 0$  in (1), we get

$$f(x)f(0) = f(x) + x.$$

We cannot have  $f(0) = 1$  since otherwise we need  $0 = x$  for any  $x \in \mathbb{R}$ , which is impossible. Then we can rewrite the above equation as  $f(x) = \frac{x}{f(0) - 1}$ . Putting  $x = 0$ , this yields  $f(0) = 0$ . Therefore,

$$f(x) = \frac{x}{0 - 1} = -x$$

for any  $x \in \mathbb{R}$ . One checks that this is a solution since

$$f(x)f(y) = (-x)(-y) = -(x+y) + xy + x + y = f(x+y) + xy + x + y.$$

**Problem 2.** Find all functions  $f : \mathbb{R} \rightarrow \mathbb{R}$  such that

$$f(xf(y)) = f(x) + xy \quad (1)$$

for any  $x, y \in \mathbb{R}$ .

**Solution.** Putting  $x = 1$  in (1), we get

$$f(f(y)) = f(1) + y. \quad (2)$$

Replacing  $x$  by  $f(x)$  in (1), and using (2), we obtain

$$f(f(x)f(y)) = f(f(x)) + f(x)y = f(1) + x + f(x)y.$$

By symmetry, we can interchange  $x$  and  $y$  so that this is equal to  $f(1) + y + f(y)x$ . This can be rewritten as

$$(f(x) - 1)y = (f(y) - 1)x. \quad (3)$$

Putting  $x = 0$  in (3), we have  $(f(0) - 1)y = 0$  for any  $y \in \mathbb{R}$ . So we must have  $f(0) = 1$ . For  $x, y \neq 0$ , the equation (3) is the same as

$$\frac{f(x) - 1}{x} = \frac{f(y) - 1}{y}.$$

This shows  $\frac{f(x) - 1}{x}$  is a constant. Therefore,  $f(x) = cx + 1$  for some constant  $c$ . Note that this also holds when  $x = 0$ . It suffices to consider functions of the form  $f(x) = cx + 1$ . Note that

$$\begin{aligned} f(xf(y)) &= f(x) + xy \\ \Leftrightarrow cx(cy + 1) + 1 &= cx + 1 + xy \\ \Leftrightarrow (c^2 - 1)xy &= 0. \end{aligned}$$

This holds for any  $x, y \in \mathbb{R}$  if and only if  $c^2 = 1$ . Therefore, the solutions are  $f(x) = x + 1$  for any  $x \in \mathbb{R}$  and  $f(x) = -x + 1$  for any  $x \in \mathbb{R}$ .

**Problem 3.** Find all functions  $f : \mathbb{Z} \rightarrow \mathbb{Z}$  such that

$$f(m - n + f(n)) = f(m) + f(n) \quad (1)$$

for any  $m, n \in \mathbb{Z}$ .

**Solution.** Suppose there exists a nonzero integer  $k$  such that  $f(k) = 0$ . By putting  $n = k$  in (1), we get  $f(m - k) = f(m)$  for any  $m \in \mathbb{Z}$ . Since  $k \neq 0$ , this shows  $f$  is periodic. In particular, as  $f$  is defined on  $\mathbb{Z}$ , it only takes finitely many possible values.

If there exists  $n \in \mathbb{Z}$  such that  $f(n) > 0$ , we can choose the largest  $M \in \mathbb{N}$  such that  $f(\ell) = M$  for some  $\ell \in \mathbb{Z}$ . By putting  $m = n = \ell$  in (1), we find that  $f(f(\ell)) = 2M$ . This contradicts the maximality of  $M$  since  $2M > M$  and  $2M$  is an image under  $f$ . Similarly, we can prove that there cannot be  $n \in \mathbb{Z}$  such that  $f(n) < 0$ . So the only possibility is  $f(n) = 0$  for all  $n \in \mathbb{Z}$ . Clearly, this is a solution.

Now, we assume  $f(n) \neq 0$  for any  $n \neq 0$ . We put  $m = 2n - f(n)$  in (1) so that  $m - n + f(n) = n$ . Therefore, the equation can be simplified to  $f(2n - f(n)) = 0$ . From the assumption, this yields  $2n - f(n) = 0$ , i.e.  $f(n) = 2n$  for any  $n \in \mathbb{Z}$ . We check that this is another solution as follows:

$$f(m - n + f(n)) = 2(m - n + 2n) = 2m + 2n = f(m) + f(n).$$

Therefore, the solutions are  $f(n) = 0$  for any  $n \in \mathbb{Z}$  or  $f(n) = 2n$  for any  $n \in \mathbb{Z}$ .

**Problem 4.** Find all functions  $f : \mathbb{R} \rightarrow \mathbb{R}$  such that

$$f(xf(y) - f(x)) = f(xy) - xy \quad (1)$$

for any  $x, y \in \mathbb{R}$ .

**Solution.** Putting  $x = 1$  in (1), we get

$$f(f(y) - f(1)) = f(y) - y. \quad (2)$$

If  $f(a) = f(b)$ , from (2) we obtain

$$a = f(a) - f(f(a) - f(1)) = f(b) - f(f(b) - f(1)) = b.$$

This implies  $f$  is injective. Then we put  $x = 0$  in (1). This yields  $f(-f(0)) = f(0)$  and hence  $-f(0) = 0$  by the injectivity. This means  $f(0) = 0$ . Next, we put  $y = 0$  in (1). Then we get  $f(-f(x)) = 0$ . Again from the injectivity, since  $f(0) = 0$ , this implies  $-f(x) = 0$ . Therefore,  $f(x) = 0$  for any  $x \in \mathbb{R}$ . This is not a solution since  $0 \neq 0 - xy$  in general. It follows that there is no solution.

**Problem 5.** Find all functions  $f : \mathbb{R} \rightarrow \mathbb{R}$  such that

$$f(x + f(x) + 2y) = 2x + 2f(f(y)) \quad (1)$$

for any  $x, y \in \mathbb{R}$ .

**Solution.** Putting  $x = y = 0$  in (1), we get  $f(f(0)) = 2f(f(0))$ . This yields

$$f(f(0)) = 0.$$

Putting  $y = 0$  in (1), we get

$$f(x + f(x)) = 2x. \quad (2)$$

This implies  $f$  is surjective since  $2x$  can be any real number. For any  $a \in \mathbb{R}$  satisfying  $f(a) = 0$ , by putting  $x = a$  in (2), we find that  $0 = 2a$ , i.e.  $a = 0$ . This shows 0 is the only real number whose image could be 0. As  $f$  is surjective, we must have  $f(0) = 0$ .

Next, we replace  $x$  by  $x + f(x)$  in (1). Using (2), this yields

$$f(x + f(x) + 2x + 2y) = 2(x + f(x)) + 2f(f(y)).$$

Note that the left hand side can be rewritten as  $f(x + f(x) + 2(x + y))$ . In view of (1), if we replace  $y$  by  $x + y$ , we obtain

$$f(x + f(x) + 2(x + y)) = 2x + 2f(f(x + y)).$$

Combining these results, we find that

$$f(f(x + y)) = f(x) + f(f(y)). \quad (3)$$

Now, for any fixed  $x \in \mathbb{R}$ , there exists  $y \in \mathbb{R}$  such that  $f(x + y) = x$  since  $f$  is surjective. For this choice of  $y$ , the equation (3) becomes  $f(f(y)) = 0$ . From above, we must have  $f(y) = 0$  and hence  $y = 0$ . Then  $f(x + y) = x$  simply means  $f(x) = x$ . This is indeed a solution since

$$f(x + f(x) + 2y) = x + x + 2y = 2x + 2f(f(y)).$$

**Problem 6.** Find all functions  $f : \mathbb{R} \rightarrow \mathbb{R}$  such that

$$f(f(x)f(y) + y^2) + f(x) = xy + f(y^2 + f(x)) \quad (1)$$

for any  $x, y \in \mathbb{R}$ .

**Solution.** Consider any  $a \in \mathbb{R}$  such that  $f(a) = 0$  (such an  $a$  may not exist). By putting  $x = a$  in (1), we get  $ay = 0$  for any  $y \in \mathbb{R}$ . So we must have  $a = 0$ . This shows 0 is the only real number whose image could be 0.

If  $f(b) = f(c)$ , then by comparing

$$\begin{cases} f(f(b)f(y) + y^2) + f(b) = by + f(y^2 + f(b)), \\ f(f(c)f(y) + y^2) + f(c) = cy + f(y^2 + f(c)), \end{cases}$$

we find that  $by = cy$  for any  $y \in \mathbb{R}$ . This shows  $b = c$ . Therefore,  $f$  is injective.

Putting  $y = \frac{f(x)}{x}$  in (1), we have

$$f\left(f(x)f\left(\frac{f(x)}{x}\right) + y^2\right) = f(y^2 + f(x)).$$

Using the injectivity, this implies  $f(x)f\left(\frac{f(x)}{x}\right) + y^2 = y^2 + f(x)$ . It follows that

$$f(x)f\left(\frac{f(x)}{x}\right) = f(x).$$

From this, we get  $f(x) = 0$  or  $f\left(\frac{f(x)}{x}\right) = 1$ . If we only consider  $x \neq 0$ , then  $f(x) \neq 0$  from above.

Therefore, we have  $f\left(\frac{f(x)}{x}\right) = 1$  for any  $x \neq 0$ . Hence, we have

$$f\left(\frac{f(x)}{x}\right) = f\left(\frac{f(y)}{y}\right)$$

for any  $x, y \neq 0$ . Again using the injectivity, we find that  $\frac{f(x)}{x} = \frac{f(y)}{y}$ . This shows  $\frac{f(x)}{x}$  is a constant. Thus,  $f(x) = cx$  for any  $x \neq 0$  where  $c$  is a constant. Note that  $c \neq 0$  since  $f$  is injective. Then  $cx$  runs through all nonzero real numbers when  $x$  varies. As  $f$  is injective, the only possible value for  $f(0)$  is 0. Hence,  $f(x) = cx$  holds for all  $x \in \mathbb{R}$ . Now,

$$\begin{aligned} & f(f(x)f(y) + y^2) + f(x) = xy + f(y^2 + f(x)) \\ \Leftrightarrow & \quad c(c^2xy + y^2) + cx = xy + c(y^2 + cx) \\ \Leftrightarrow & \quad (c^3 - 1)xy + (c - c^2)x = 0. \end{aligned}$$

This holds for any  $x, y \in \mathbb{R}$  if and only if  $c^3 - 1 = c - c^2 = 0$ , i.e.  $c = 1$ . Thus,  $f(x) = x$  is the only solution.

**Problem 7.** Find all functions  $f : \mathbb{N} \rightarrow \mathbb{N}$  such that

$$f(m+n) + n \mid m + f(n) \quad (1)$$

for any  $m, n \in \mathbb{N}$ .

**Solution 1.** Putting  $m = 1$  in (1), we get  $f(1+n) + n \mid 1 + f(n)$ . Since both sides are positive, this implies

$$f(1+n) + n \leq 1 + f(n).$$

For  $n > 1$ , this yields  $f(n+1) < f(n)$ . Therefore, the sequence  $f(2), f(3), \dots$  is a strictly decreasing sequence of positive integers. This is impossible. Hence, there is no such function  $f$ .

**Solution 2.** We choose any prime  $p > f(1)$ . Putting  $m = p - f(1)$  and  $n = 1$  in (1), we get

$$f(p - f(1) + 1) + 1 \mid p.$$

Since  $p$  is a prime and  $f(p - f(1) + 1) + 1 > 1$ , we must have  $f(p - f(1) + 1) + 1 = p$ , i.e.

$$f(p - f(1) + 1) = p - 1$$

for any prime  $p > f(1)$ . Next, for any  $n \in \mathbb{N}$  and any prime  $p > \max\{f(1), f(1) - 1 + n\}$ , we put  $m = p - f(1) + 1 - n$  in (1). Then we have

$$p - 1 + n \mid p - f(1) + 1 - n + f(n).$$

This yields

$$p - 1 + n \mid -f(1) + 2 - 2n + f(n).$$

If we fix  $n \in \mathbb{N}$  and choose a sufficiently large  $p$ , then the left-hand side is larger than the absolute value of the right-hand side. Therefore, the only possibility is  $-f(1) + 2 - 2n + f(n) = 0$ . In other words,  $f(n) = 2n + c$  for some constant  $c \in \mathbb{Z}$ . Putting this in (1), we find that

$$\begin{aligned} & f(m+n) + n \mid m + f(n) \\ \Rightarrow & 2(m+n) + c + n \mid m + 2n + c \\ \Rightarrow & 2m + 3n + c \mid m + 2n + c. \end{aligned}$$

If we choose  $m = n$ , this yields  $5n + c \mid 3n + c$ , which is impossible since  $5n + c > 3n + c > 0$  for sufficiently large  $n$ . Therefore, there is no solution.

**Problem 8.** (Iran MO 3rd Round 2018 Number Theory Problem 3) Find all functions  $f : \mathbb{N} \rightarrow \mathbb{N}$  so that for every natural numbers  $m, n$ ,  $f(n) + 2mn + f(m)$  is a perfect square.

**Solution.** Let

$$f(n) + 2mn + f(m) = a^2, \quad (1)$$

$$f(n) + 2(m+1)n + f(m+1) = b^2, \quad (2)$$

$$f(n) + 2(m+2)n + f(m+2) = c^2 \quad (3)$$

for some  $a, b, c \in \mathbb{N}$ . Taking the difference of (1) and (2), we get

$$2n + f(m+1) - f(m) = b^2 - a^2. \quad (4)$$

We claim that  $f(m+1) - f(m)$  is odd. Indeed, if it is even, then  $2 \mid b^2 - a^2$ . It is well-known and easy to check that  $b^2 - a^2 \not\equiv 2 \pmod{4}$ . So we must have  $4 \mid b^2 - a^2$ . Hence  $4 \mid 2n + f(m+1) - f(m)$ . However, we can use the same arguments to show that  $4 \mid 2(n+1) + f(m+1) - f(m)$ . The two divisibilities cannot hold simultaneously. Therefore,  $f(m+1) - f(m)$  is odd.

For any fixed  $m \in \mathbb{N}$ , we can choose  $n = \frac{p - f(m+1) + f(m)}{2}$  for some large odd prime  $p$ . Then  $2n + f(m+1) - f(m) = p$ . From (4), we have  $p = (b-a)(b+a)$ . Thus,  $b-a=1$  and  $b+a=p$ . This yields

$$a = \frac{p-1}{2} \quad \text{and} \quad b = \frac{p+1}{2}.$$

Next, taking the difference of (2) and (3), we get

$$2n + f(m+2) - f(m+1) = c^2 - b^2. \quad (5)$$

If we further take the difference of (4) and (5), we obtain

$$f(m+2) - 2f(m+1) + f(m) = c^2 - 2b^2 + a^2.$$

Noting that  $2b^2 - a^2 = 2\left(\frac{p+1}{2}\right)^2 - \left(\frac{p-1}{2}\right)^2 = \frac{p^2 + 6p + 1}{4}$ , we can rewrite the above equation as

$$c^2 = d + \frac{p^2 + 6p + 1}{4}$$

where  $d = f(m+2) - 2f(m+1) + f(m)$  is fixed given that  $m$  is fixed. For sufficiently large prime  $p$ , we have

$$\left(\frac{p+1}{2}\right)^2 < d + \frac{p^2 + 6p + 1}{4} < \left(\frac{p+5}{2}\right)^2.$$

Since the middle term is  $c^2$ , the only possibility is

$$d + \frac{p^2 + 6p + 1}{4} = \left(\frac{p+3}{2}\right)^2.$$

This simplifies to give

$$f(m+2) - 2f(m+1) + f(m) = d = 2.$$

We have

$$\begin{aligned} f(3) &= 2f(2) - f(1) + 2, \\ f(4) &= 2f(3) - f(2) + 2 = 3f(2) - 2f(1) + 6, \\ f(5) &= 2f(4) - f(3) + 2 = 4f(2) - 3f(1) + 12, \\ &\dots \end{aligned}$$

By induction, it is easy to deduce

$$f(n) = (n-1)f(2) - (n-2)f(1) + (n-1)(n-2)$$

for any  $n \in \mathbb{N}$ . In other words,  $f(n)$  is a monic quadratic polynomial in  $n$ . For simplicity, we write

$$f(n) = n^2 + kn + \ell$$

for some integer constants  $k, \ell$ . When  $m = 1$ , the given condition says that

$$f(n) + 2n + f(1) = n^2 + (k+2)n + (k+2\ell+1)$$

is a square number. This shows

$$4n^2 + 4(k+2)n + 4(k+2\ell+1) = (2n+k+2)^2 + (8\ell-k^2)$$

is a square number. For sufficiently large  $n$ , we have

$$(2n+k+1)^2 < (2n+k+2)^2 + (8\ell-k^2) < (2n+k+3)^2.$$

Therefore, we need  $(2n+k+2)^2 + (8\ell-k^2) = (2n+k+2)^2$ , i.e.  $8\ell = k^2$ . This shows  $4 \mid k$  and hence we can write  $k = 4t$  and  $\ell = 2t^2$  for some  $t \in \mathbb{Z}$ . Then we have

$$f(n) = n^2 + 4tn + 2t^2. \tag{6}$$

If  $t < 0$ , then we put  $n = -t$  to get  $f(n) = -t^2 < 0$ . This is impossible. So we have  $t \geq 0$ . We claim that all such functions are solutions. Firstly, for  $t \geq 0$ , we must have  $f(n) > 0$  in view of (6). It remains to check

$$f(n) + 2mn + f(m) = (n^2 + 4tn + 2t^2) + 2mn + (m^2 + 4tm + 2t^2) = (m+n+2t)^2$$

is a perfect square. Therefore, the solutions are given by (6) where  $t$  is a nonnegative integer.